INCH-POUND

MIL-DTL-14652B
18 December 2001
SUPERSEDING
MIL-I-14652A
13 March 1992

DETAIL SPECIFICATION

INSIGNIA, EMBROIDERED, ORGANIZATIONAL, BADGES, AND BRANCH OF SERVICE, ARMY AND AIR FORCE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-14652 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

1. SCOPE

1.1 Scope. This specification establishes the requirements for embroidered organizational insignia, badges, and branch of service insignia.

1.2 Classification. The insignia covered by this specification will be of the following types, styles, and classes as specified (see 6.2):

Type I - Organizational Insignia
- Style 1 - Army
- Style 2 - Air Force
- Class 1 - Full Color
- Class 2 - Woodland Subdued
- Class 3 - Desert Subdued

Type II - Badges
- Style 1 - Army
- Style 2 - Air Force
- Class 2 - Woodland Subdued
- Class 3 - Desert Subdued

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in preparing this document should be addressed to: Director, The Institute of Heraldry, US Army, 9325 Gunston Road, Room S112, Fort Belvoir, VA 22060-5579, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

AMSC N/A FSC 8455

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Type III - Insignia, Branch of Service
Style 1 - Army
Style 2 - Air Force
Class 2 - Woodland Subdued
Class 3 - Desert Subdued

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

FEDERAL SPECIFICATIONS

A-A-50199 - Thread, Polyester Core, Cotton- or Polyester- Covered
A-A-52094 - Thread; Cotton

FEDERAL STANDARDS

FED-STD-191 - Textile, Test Methods

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.) Or visit the DODSSP website at http://astimage.daps.dla.mil for online access to standardization documents.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

THE INSTITUTE OF HERALDRY DRAWINGS AND CARTOONS

A-1 (Series) Army Shoulder Sleeve Insignia, Various
A-4 (Series) Beret Flash, Various
A-6 (Series) Background Trimmings, Various
A-10J (Series) Shoulder Sleeve Insignia, JROTC, Various
A-10S (Series) Shoulder Sleeve Insignia, SROTC, Various
A-12 (Series) Patches, USAF, Squadrons, Various
A-12S (Series) Patches, USAF, Various

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A-13 (Series) Badges, Patches, Identification, Various
B-2 (Series) Insignia, Branch of Service, Various
B-5 (Series) Badges, Qualification, Occupational, Various
B-15 (Series) USAF Command/Organizational Patches, Various

(Copies of specifications, standards, drawings, and specification sheets required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are issues of the documents cited in the solicitation (see 6.2)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

AATCC-16 - Colorfastness to Light
AATCC-61 -Colorfastness to Laundering, Home and Commercial; Accelerated
AATCC-96 - Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool
AATCC-99 - Dimensional Changes of Woven or Knitted Wool Textiles: Relaxation, Consolidation

(These standards can be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

AMERICAN SOCIETY FOR QUALITY CONTROL

ANSI/ASQC Z1.4 1993 - Sampling Procedures and Tables for Inspection by Attributes

(This standard can be obtained from the American Society for Quality Control, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D276 - Standard Test Methods for Identification of Fibers in Textiles
ASTM-D1907 - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method
ASTM-D3775 - Standard Test Method for Fabric Count of Woven Fabric
ASTM-D3776 - Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM-D5034 - Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Grab Test)

(These standards can be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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CARABELA TRADING COMPANY

C-Series Base Cloth Shades

(Information on the referenced base cloth shades can be requested from Carabela Trading Company, 10 Fairmant Avenue, Chatham, New Jersey 07928.)

CARR TEXTILES

PC-Series Base Cloth Shades

(Information on the referenced base cloth shades can be requested from Carr Textiles, 243 Wolfner Drive, Fenton, Missouri 63026.)

IRA GREEN, INC.

DOD Standard Shades for Heraldic Yarns for Embroidery and Ribbons, 1993 Edition

(This publication can be obtained from Ira Green, Inc., 177 Georgia Avenue, Providence, RI 02905.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the materials meet or exceed the operational and maintenance requirements, and promote economically advantageous life cycle costs.

3.3 Materials. Materials shall be as specified herein. Verification shall be in accordance with the standards and methods listed in 4.3.1.

3.3.1 Embroidery yarn.

3.3.1.1 Material identification. The embroidery yarn shall be a semi-dull continuous filament, regenerated cellulose type rayon, 2-ply Schiffli type or Multihead yarn. The use of 2-ply polyester is authorized in lieu of rayon. Denier of the yarn shall be as specified on the drawing or cartoon. Comparable sizes are as follows:

SCHIFFLI		**MULTIHEAD**	
Rayon	Polyester	Rayon	Polyester
150/2	135/2	150/2 or 35 weight	-----
100/2	90/2	120/2 or 40 weight	120/2 or 40 weight
75/2	70/2	-----	95/2

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3.3.1.2 Color. The color shall be as specified on the applicable drawing or cartoon. When overedge stitching is required, the color shall be the same as the embroidered border. The use of sulfur dyes is prohibited.

3.3.1.3 Matching. The color shall match the specified color from the DOD Standard Shades for Heraldic Yarns for Embroidery and Ribbons, 1993 Edition, when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7000 ± 500 K, with illumination of 100 ± 20 foot candles. The color shall also be a good match to the specified color under incandescent lamplight at 2300 ± 100 K. The degree of luster shall be equal to the standards established in the DOD Standard Shades for Heraldic Yarns for Embroidery and Ribbons, 1993 Edition.

3.3.1.4 Colorfastness. When tested, the dyed embroidered yarn shall show good fastness to wet-dry cleaning, laundering, and light except the colorfastness for black, blue, and red shades shall be not less than fair for wet dry-cleaning and laundering.

3.3.2 Bobbin yarn. The bobbin yarn shall be a continuous multi-filament nylon, 2-ply cotton, or 2-ply spun polyester yarn. The yarn shall be undyed except that the yarn for Types II and III insignia may be dyed black. When the bobbin yarn is dyed black, it shall show fair fastness to wet-dry cleaning and laundering. The use of sulfur dyes is prohibited.

3.3.3. Overedge stitching yarn and thread.

3.3.3.1 Material identification. The yarn and thread used for overedge stitching (see 3.5.3.2) shall be 600 or 1200 denier multi-filament bright rayon or polyester for cover thread and Type III, Ticket No. 00, 3-ply of A-A-52094 or Ticket No. 50, 2 ply of A-A-50199 for needle and looper threads.

3.3.3.2 Color. The color of the yarn and the thread used for overedge stitching shall be as specified for the embroidered border on the applicable drawing or cartoon. The use of sulfur dyes is prohibited.

3.3.3.3 Colorfastness. When tested, the dyed overedge stitching yarn and thread shall show good fastness to wet-dry cleaning, laundering, and light except the colorfastness for black, blue, and red shades shall be not less than fair for laundering.

3.3.4 Table I. Table I shows the color combinations of embroidery and overedge yarns that are authorized for use on the face of each type, style, and class of insignia.

TABLE I. **Color of embroidery and overedge yarns.**

Type, Style, Class	**Yarn Colors**
Type I,	
Style 1, Class 1	As specified on cartoon or drawing
Style 1, Class 2	Two or more of the following colors only: Black HAN 67138, Olive Drab HAH 67133, Brown HAL 67136
Style 1, Class 3	Two or more of the following colors only: Spicebrown HCV 67196, Khaki HCS 67193, Brown HAL 67136
Style 2, Class 1	As specified on cartoon or drawing

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TABLE I. **Color of embroidery and overedge yarns.** - Continued

Type, Style, Class	Yarn Colors
Style 2, Class 2	Two or more of the following colors only: Black HAN 67138, Olive Drab HAH 67133, Garnet HBH 67158, Flag Blue HY 67124, Spruce Green HAE 67130
Style 2, Class 3	Two or more of the following colors only: Spicebrown HCV 67196, Khaki HCS 67193, Black HAN 67138, Brown HAL 67136
Type II	
Style 1, Class 2	One or more of the following colors only: Black HAN 67138, Brown HAL 67136, Olive Drab HAH 67133, Spruce Green HAE 67130
Style 1, Class 3	One or more of the following colors only: Spicebrown HCV 67196, Brown HAL 67136, Khaki HCS 67193
Style 2, Class 2	One or more of the following colors only: Flag Blue HY 67124, Olive Drab HAH 67133
Style 2, Class 3	One or more of the following colors only: Khaki HCS 67193, Black HAN 67138 or Spicebrown HCV 67196
Type III	
Style 1, Class 2	One or more of the following colors only: Black HAN 67138, Brown HAL 67136, Olive Drab HAH 67133, Spruce Green HAE 67130
Style 1, Class 3	One or more of the following colors only: Spicebrown HCV 67196, Brown HAL 67136, Khaki HCS 67193
Style 2, Class 2	One or more of the following colors only: Flag Blue HY 67124, Olive Drab HAH 67133
Style 2, Class 3	One or more of the following colors only: Khaki HCS 67193, Black HAN 67138 or Spicebrown HCV 67196

3.3.5 Base cloth. Base cloth for all insignia shall be 100% cotton, 50/50 polyester/cotton, or 65/35 polyester/cotton, three up, one down, left hand twill, 7.0 to 8.3 oz. per square yard, single ply warp and filling, with a breaking strength of not less than 150 lbs. in the warp and 80 lbs. in the filling. The residual shrinkage shall be not more than 2 % in either the warp or filling direction.

3.3.5.1 Color. The color of the base cloth shall be as shown in Table II. ✳DPSC Handbook 4120-1 color references are from the *List of Armed Services Colors for Clothing, Textiles, and Equipment, 31 March 1989,* which is obsolete. The DPSC colors are included for historical reference only. Base cloth colors should be the same as or equal to the Carabela or Carr colors cited. The cloth shall have not less than fair colorfastness to laundering when tested.

TABLE II. **Color of base cloth.**

Type, Style, Class	Base Cloth Color		
	✳DPSC Handbook	Carabela	Carr
Type I			
Style 1, Class 1	Khaki 2120	C4120	PC-TAN
Style 1, Class 2	Olive Green 107	C2012	PC-OLV

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TABLE II. **Color of base cloth.** - Continued

Type, Style, Class	Base Cloth Color		
	✳DPSC Handbook	Carabela	Carr
Style 1, Class 3	Khaki 2120	C4120	PC-TAN
Style 2, Class 1	Khaki 2120	C4120	PC-TAN
Style 2, Class 2	Olive Green 107	C2012	PC-OLV
Style 3, Class 3	Khaki 2120	C4120	PC-TAN
Type II			
Style 1, Class 2	Olive Green 107	C2012	PC-OLV
Style 1, Class 3	Khaki 2120	C4120	PC-TAN
Style 2, Class 2	Olive Green 107	C2012	PC-OLV
Style 2, Class 3	Khaki 2120	C4120	PC-TAN
Type III			
Style 1, Class 2	Olive Green 107	C2012	PC-OLV
Style 1, Class 3	Khaki 2120	C4120	PC-TAN
Style 2, Class 2	Olive Green 107	C2012	PC-OLV
Style 2, Class 3	Khaki 2120	C4120	PC-TAN

3.3.6 Backing material. The backing material, when specified, shall be a light, open-weave, single ply fabric weighing 2.5 oz. (minimum) per square yard and having a minimum thread count of 40 ends in the warp and 20 ends in the filling. It shall have a minimum breaking strength of 30 pounds in the warp direction and 20 pounds in the filling direction. At the option of the manufacturer, a non-woven pressed fiber material shall be permitted as backing material provided that it satisfies the performance requirements stated herein.

3.3.7 Stiffener. When specified, insignia shall be treated on the reverse side with a fabric finish which will stiffen the insignia to facilitate subsequent processing. The fabric finish shall not affect the colors on the face of the insignia; cause puckering which seriously affects the appearance of insignia; or react adversely to storage, laundering, wet-dry cleaning, or ironing.

3.4 Design. The design and color placement for embroidered insignia shall be as shown on the Government furnished cartoon or drawing (see 6.5).

3.5 Construction. The insignia shall be embroidered on an automatic type embroidery machine in accordance with this specification and the Government furnished cartoon. Cartoons are furnished for design; color of yarns; placement of stitches; number of under, cover, and miscellaneous stitches; and minimum stitches required. Because all cartoons are based on Schiffli embroidery machine capabilities, the stitch count requirements required by the applicable cartoon shall be doubled for other than Schiffli embroidery machines. The density of cover stitches may be increased in critical areas where defects may occur. All thread ends shall be trimmed.

3.5.1 Embroidery stitches. The embroidery stitches shall be of the type, length, number, and density specified on the applicable Government cartoon. The total stitches specified on the cartoon are full stitches and constitute two revolutions or movements of the Schiffli machine. Underlay stitches shown on the cartoon by numbers are 1/2 stitches (one movement of the machine) but are converted to full stitches on the tabular listing. Except where otherwise

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indicated in the cartoon, all letters shall be underlaid and formed with blatt stitches. When required by contract provisions, the contractor shall submit a sample for approval. When a first article has been specifically approved for appearance of the embroidery stitching prior to production, deliveries shall be in this respect equal to or better than the approved article.

3.5.1.1 Background stitches. Insignia requiring background stitching shall have either horizontal or vertical background stitching (as indicated on the applicable Government cartoon), and the stitch shall be 3/16 inch in length ± 1/32 inch interlocking in evenly spaced rows (satin fill). Horizontal background stitching shall run in a vertical row and vertical background stitching shall run in a horizontal row. The visible width of the embroidered row shall not be more than 5/64 inch. The number of visual rows and the density (number of stitches per inch in a stitch line) shall be as indicated on the applicable Government cartoon. When a Government cartoon or drawing is not available, all Type I, Styles 1 and 2 insignia shall have an embroidered background.

3.5.1.2 Device stitches. Device stitches shall be in accordance with the applicable Government cartoon with respect to spacing of the rows, number of visible rows, and length and density of stitches. Translation of curvatures and other device features shall be executed with embroidery stitching finished evenly to effect a clean outline.

3.5.1.3 Underlay stitches. When underlay stitches are required, the number of stitches shall be as indicated on the cartoon or drawing. Where underlay is required in areas of design showing rows of stitches, the underlay stitches shall be inserted in a criss-cross pattern. In other areas requiring underlay, the underlay stitches shall be inserted in the direction most suitable for proper design execution.

3.5.1.4 Stiele stitches. The length and density of required stiele stitches shall be tightly drawn and well executed to produce a clear, sharp edging and shall conform to the applicable Government cartoon.

3.5.1.5 Border stitches. All Type I insignia shall be bordered by a schrage stitch or a combination of a blatt stitch with a stiele stitch, provided it is equal in density to the specified schrage stitch.

3.5.2 Stiffener. Type I insignia may be treated on the reverse side with a fabric finish such as that described in 3.3.7. The use of backing material specified in 3.3.6 shall be permitted provided that it does not protrude at any point beyond the outline of the finished insignia.

3.5.3 Cutting and finishing.

3.5.3.1 Cutting.

3.5.3.1.1 Type I. Type I insignia shall be cut in single units, and the base cloth shall be evenly trimmed as close to the outside of the embroidered edge as is possible without cutting the stitching. For Type I, Style 1 insignia, the base cloth shall be trimmed close enough to ensure the overedge stitching covers the embroidered border when required.

3.5.3.1.2 Types II and III. Types II and III insignia shall be cut in single units with a minimum of 5/8 inch base material is available between the design and the raw edge.

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3.5.3.2 Overedge stitching. Overedge stitching shall be superimposed over the border of all Type I, Style 1 insignia using yarn and thread specified in 3.3.3 as follows:

a. Overedge stitching shall conform to stitch Type 504 (Figure 37) or 505 (Figure 38) of ASTM D6193 except two ends of 600 denier or one end of 1200 denier shall be used for the cover threads. These shall be 24 stitches ± 4 stitches per inch, and the stitch length shall be 1/8 inch ± 1/32 inch. Tolerance on width shall be acceptable only if the overedge stitching completely covers the border stitching when the specified border on the drawing is 1/8 inch in width.

b. On circular or oval designs, there shall be a 1/2 inch overlap of the overedge stitching centered on the bottom of the insignia.

c. The overedge stitching shall start and finish at the bottom point for pointed shields.

d. On rounded shields or similar designs with the bottom of the insignia round, the overedge shall start and finish on the upper left corner.

e. Stitching for triangular, square, rectangular, polygonal, cylindrical, or barrel shaped designs shall start and finish on any corner.

f. Where practicable, overedge stitching shall be continuous from start to finish.

g. Unless shown on the cartoon or drawing, the rounding of corners or angles shall not be permitted.

h. If stitching runs off the edge to begin a new direction for corners, the stitching ends shall be trimmed before being locked in by overedge stitching in the new direction.

i. The final free end of the overedge stitching shall be brought to the back of the insignia and secured with adhesive. The free end shall not be less than 1/2 inch in length, and no adhesive shall be permitted within 1/8 inch of the finished insignia edge.

3.5.3.2.2 Optional methods of securing final free end. At the option of the contractor, the final free end of the overedge stitching may secured by blind-stitching 1/4 inch or less of the terminal end to the back of the insignia by hand or machine. Or where size and configuration permit, the final end may be secured by application of a 3/8 inch by 1 inch (approximate) rectangle of masking tape. No part of the tape shall be within 1/8 inch of the finished edge of the insignia.

3.6 Tolerances. Unless otherwise specified, the overall measurements of the finished insignia shall be within a tolerance of ± 1/8 inch of the required measurements. Where no tolerances are cited, design dimensions shall have a tolerance of ± 1/32 inch. Examination shall be in accordance with 4.3.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection. Inspection and testing of the first article (see 3.1) shall be made of a completely finished item or items for all provisions of this specification applicable to the end product examination and tests.

b. Conformance inspection. Inspection shall be in accordance with the provisions set forth herein, except where otherwise indicated (see 6.2). Sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in 4.3.

4.3 Methods of inspection.

4.3.1 Testing of components. Except as otherwise stated in TABLE III, testing of components shall be in accordance with the stated test method from FED-STD-191.

TABLE III. **Testing of components.**

COMPONENT	CHARACTERISTIC	RQM'T ¶	TEST METHOD	RESULTS REPORTED AS
Embroidery yarn	material identification	3.3.1.1	ASTM-D276; 1/	pass or fail
	ply	3.3.1.1	visual	pass or fail
	denier	3.3.1.1	ASTM-D1907	to nearest denier
	color	3.3.12 & 3.3.1.3	visual	pass or fail
	presence of sulfur	3.3.1.2	2/	pass or fail
	colorfastness	3.3.1.4	5622	pass or fail
Bobbin yarn	material identification	3.3.2	ASTM-D276; 1/	pass or fail
	ply	3.3.2	visual	pass or fail
	color	3.3.2	visual	pass or fail
	presence of sulfur	3.3.2	2/	pass or fail
	colorfastness	3.3.2	5622	pass or fail
Overedge stitching yarn	material identification	3.3.3.1	ASTM-D276; 1/	pass or fail
	ply	3.3.3.1	visual	pass or fail
	denier	3.3.3.1	ASTM-D1907	to nearest denier
	color	3.3.3.2	visual	pass or fail
	presence of sulfur	3.3.3.2	2/	pass or fail
	colorfastness	3.3.3.3	5622	pass or fail
Base cloth	material identification	3.3.5	ASTM-D276; 1/	pass or fail
	weave	3.3.5	visual	pass or fail
	weight	3.3.5	ASTM-D3776, Option A	nearest 0.1 oz/ sq yd
	ply	3.3.5	visual	pass or fail
	breaking strength: warp & filling	3.3.5	ASTM-D-5034, Tests G-E	nearest 1 lb.
	shrinkage: warp & filling	3.3.5	AATCC-96-Test Vc, D & AATCC-99	nearest 0.1%
	colorfastness	3.3.5.1	5622	pass or fail
	color	3.3.5.1	visual	pass or fail

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TABLE III. **Testing of components.** - Continued

COMPONENT	CHARACTERISTIC	RQM'T ¶	TEST METHOD	RESULTS REPORTED AS
Backing material	weave	3.3.6	visual	pass or fail
	weight	3.3.6	ASTM-D3776, Option A	nearest 0.1 oz/ sq yd
	thread count: warp & filling	3.3.6	ASTM-D3775	pass or fail
	breaking strength: warp & filling	3.3.6	ASTM-D5034, Test G-E	nearest 1 lb.

1/ Unless otherwise specified, a contractor's certification of compliance may be accepted as evidence of conformance.

2/ Accepted on contractor's certification of compliance.

4.3.2 Visual examination of the end item. Visual examination of the completely finished item shall be made using the guidelines given in Table IV. Examination shall be made at a distance of approximately 16-22 inches with illumination equal to average daylight and arranged so as to avoid as much reflected light as possible. Defects designated by an asterisk (*) shall be classified as major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously.

TABLE IV **Visual examination of the end item.**

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(*)	MINOR
Base cloth	Other than specified.	X		
	Any splice, cut, hole or tear visible on face.	X		
	Color/shade not as specified.	X		
Stiffener/Backing	Backing material omitted or incomplete when required.	X		
	Coating not uniform; incomplete.			X
	Backing material protrudes more than 1/32 inch beyond finished insignia.			X
	Stiffener visible on front.	X		
	Stiffener/backing not as specified.	X		
Embroidery, General	Embroidery yarn not an acceptable match.	X		
	Background not as specified.	X		
	Device not as specified.	X		
	Background not interlocked in evenly spaced rows	X		
	Spacing of rows not in accordance with cartoon. .		*	
	Non-uniform embroidery stitch density.		*	
	Bobbin thread shows on face.		*	
	Tight tension; distorted surface of face side.	X		
	Loose tension; loopy surface.	X		
	Traveling (walking) stitches on face of insignia. . .		*	
	Broken stitches or stitching incomplete (missing). .		*	

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TABLE IV **Visual examination of the end item.** - Continued

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(*)	MINOR
Design	Any departure from cartoon or drawing.	X		
	Design or device poorly defined, sharp outline lacking.	X		
	Shift; device or outline misshapen.		*	
	Device flat; underlay missing or not well rounded or prominent; underlay insufficient.	X		
	Design inadequately covers base cloth; background or underlay showing through (thin or uneven stitch density).	X		
	Border not constructed of schrage stitches or a combination of blatt and stiele stitches when required.	X		
	Stiele stitches not firmly drawn to produce clearly defined rows or edges.	X		
	Stiele stitch in device not as required by cartoon. .	X		
Workmanship, General	Kinked, bunched, tangled yarn on face.		*	
	Broken, scuffed, or cut yarns on face.	X		
	Spot, stain, stiffener, or foreign matter on face. . . .		*	
	Two or more thread ends untrimmed.			X
	Creased or folded; will not flatten by manual pressure.		*	
Finishing	Type I – Embroidered Borders & Overedge			
	Omitted completely or not as specified.	X		
	Final free end not secured on back of insignia as specified.		*	
	Final free end not specified length.			X
	Overedge does not adequately cover border.	X		
	Bunched or tangled yarns.		*	
	Broken or skipped stitches.		*	
	Raveled; ends not secured.		*	
	Run-off; not caught in base cloth.		*	
	Not 20-28 stitches per inch on overedge.		*	
	Start and finish not at specified point.		*	
	Angles or pointed corners rounded off.	X		
	Too tight or too loose tension.			X
	Overedge less than 3/32 or more than 5/32 inch in length.		*	
	Adhesive from final end exposed on finished edge		*	
	Cut shrage or border stitches not adequately covered by overedge stitching.	X		
	Wrong stitch type.	X		
	Not cut in single units as required.			

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TABLE IV **Visual examination of the end item.** - Continued

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(✳)	MINOR
	Types II and III – Base Cloth Borders and Edges Not uniform in width or unevenly trimmed. . . . . . Corners or points not well defined. . . . . . Free material between edge of design and raw edge of material not as specified. . . . . . Not cut in single units when required.	X	✳ ✳	

4.3.3 Examination for dimensions. Measurements shall be taken of all points, lines, areas, curves, radii, and spaces for which dimensions are specified in the cartoon or drawing, including:

Overall length
Overall width
Border width
Device location
Device dimension
Spaces and distances between intermediate points

Tolerances shall be as specified in 3.6. Any dimension that is not within the specified tolerance shall be classified as a major defect.

4.3.4 End item testing. Except as otherwise stated in TABLE V, testing of the completely finished item(s) shall be in accordance with FED-STD-191. "The completely finished item" is defined as one insignia.

TABLE V. **End item testing.**

CHARACTERISTIC	RQM'T ¶	TEST METHOD
Colorfastness of embroidery & overedge stitching yarn to:		
Wet-dry cleaning	3.3.1.4 & 3.3.3.3	5622
Light	3.3.1.4 & 3.3.3.3	AATCC-16, Option A
Laundering **1**/	3.3.1.4 & 3.3.3.3	AATCC-61, Test 3A
Overedge stitching: Stitches per inch	3.5.3.2	5052 **2**/

1/ Only the stain on the fiber of the color transfer cloth represented in the basic materials being tested, shall be evaluated.

2/ The contractor shall furnish a certificate of compliance.

5. PACKAGING

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5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The insignia covered by this specification are intended to be worn by Department of Defense personnel to designate wearers' organization, branch of service, or skill.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, style, and class required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).
d. Drawing/cartoon number (see 2.2.2).
e. When a first article is not required (see 3.1).
f. Packaging requirements (see 5.1 and 6.6).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When first article inspection is required (see 3.1), the contracting office should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first production sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Government furnished property. The contracting office should arrange to furnish the cartoons and drawings required in 3.3, 3.4, and 3.5.

6.6 Suggested packaging. Packaging, packing, preservation, marking, and any acceptance criteria will be in accordance with 5.1. Suggested packaging is given below. Please note that acceptable packaging may take a number of different forms commercial wrapping paper; clear blister or "bubble" wrap; shrink wrap; bags made of commercial grade paper, clear plastic, or clear blister/"bubble" pack closed by tying, taping, or heat-sealing, as appropriate. The

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intention is to keep a stated number of units of issue together for counting, marking, and intermediate packaging (if any) and to do so in a manner that does no damage to the insignia during handling, shipping, and storing.

Insignia of one type, style, and class may be put up in bundles of twenty each.

6.7 Subject term (key word) listing.

Bobbin yarn
Cartoon
Colorfastness
Denier
Multihead
Schiffli
Stiffener
Stitch, background
Stitch, blatt
Stitch, blind
Stitch, border
Stitch, overedge
Stitch, schrage
Stitch, stiele
Stitch, underlay
Sulfur

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - IH
Air Force - 11

Preparing activity:
Army - IH

Review activities:
Air Force - 99
DLA - CT

(Project No. 8455-0070/001)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-14652B	2. DOCUMENT DATE (YYYYMMDD) 20011218

3. DOCUMENT TITLE INSIGNIA, EMBROIDERED, ORGANIZATIONAL, BADGES, AND BRANCH OF SERVICE, ARMY AND AIR FORCE, GENERAL SPECIFICATION FOR

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

NAME The Institute of Heraldry, Dept. of the Army	b. TELEPHONE *Include Area Code)* (1) Commercial 703-806-4982 (2) AUTOVON 656-4982
c. ADDRESS *(Include Zip Code)* Technical and Production Division 9325 Gunston Rd., Bldg. 1466, Rm S-112 Fort Belvoir, Virginia 22060-5579	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533, Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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